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Comparative Genomics

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Bacterial Genomic Surveillance
June 2026

PLOS | BIOLOGY

PERSPECTIVE

Big Data: Astronomical or Genomical?

Abstract

Genomics is a Big Data science and is going to get much bigger, very soon, but it is not known whether the needs of genomics will exceed other Big Data domains. Projecting to the year 2025, we compared genomics with three other major generators of Big Data: astronomy, YouTube, and Twitter. Our estimates show that genomics is a “four-headed beast”—it is either on par with or the most demanding of the domains analyzed here in terms of data acquisition, storage, distribution, and analysis. We discuss aspects of new technologies that will need to be developed to rise up and meet the computational challenges that genomics poses for the near future. Now is the time for concerted, community-wide planning for the “genomical” challenges of the next decade.

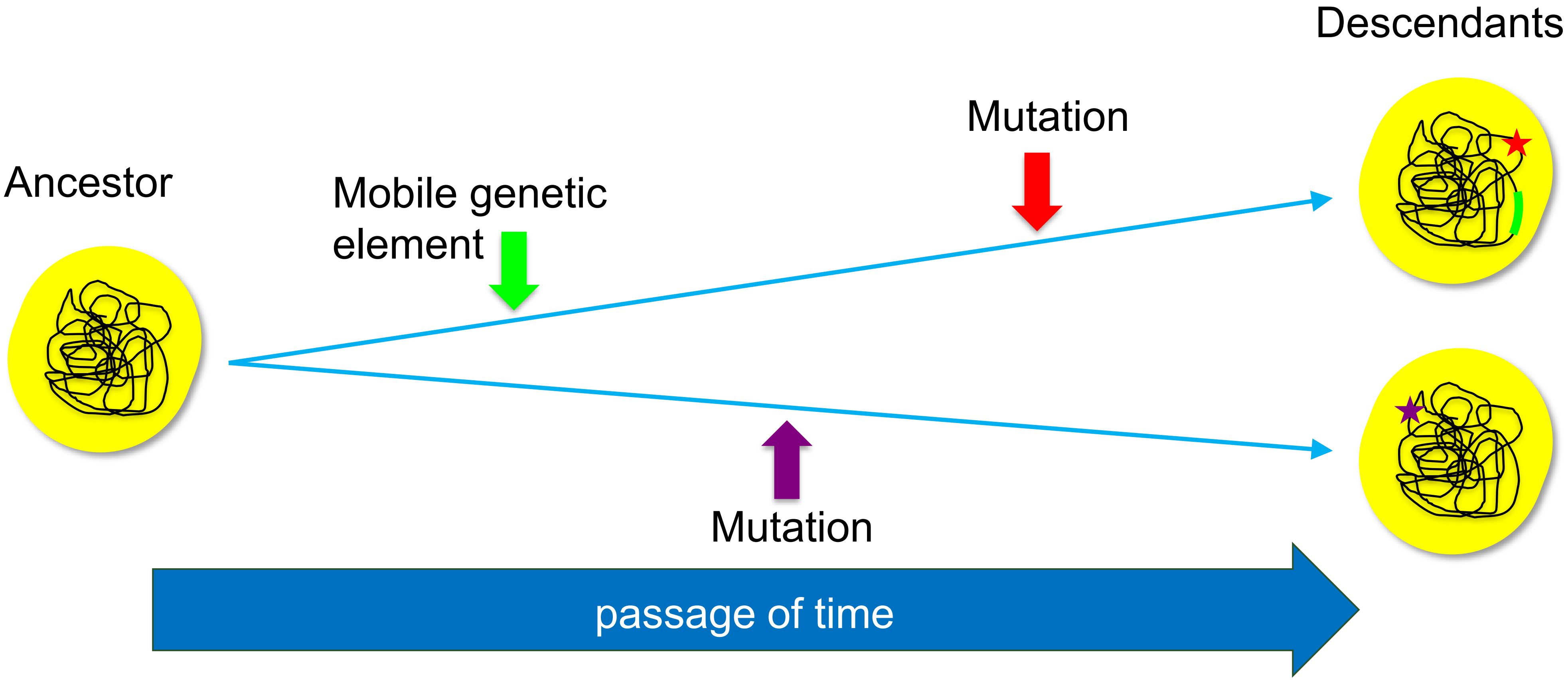
Stephens et al. PLoS Biol. 2015. doi: 10.1371/journal.pbio.1002195.

Genomes are genetic inventories

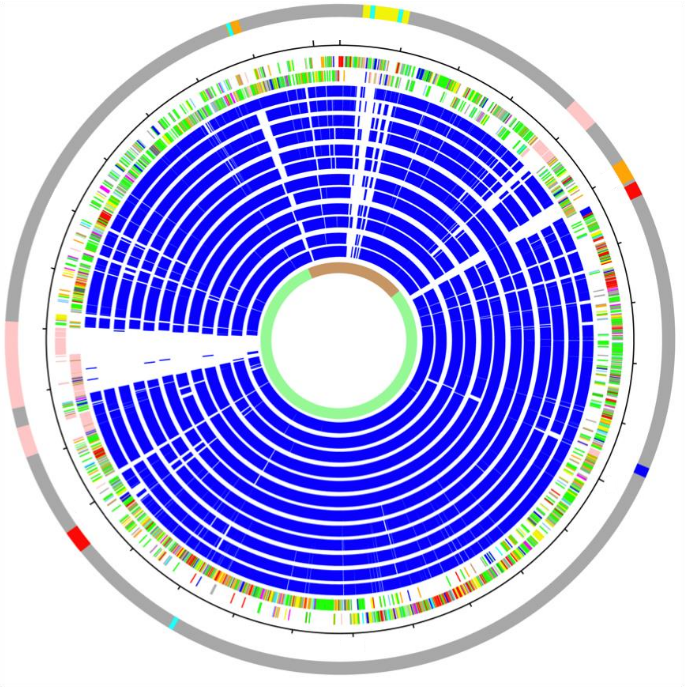
Relationship of Genotype - Phenotype

Genes that encodes virulence factors, antibiotic resistance determinants

Genomes are historical records



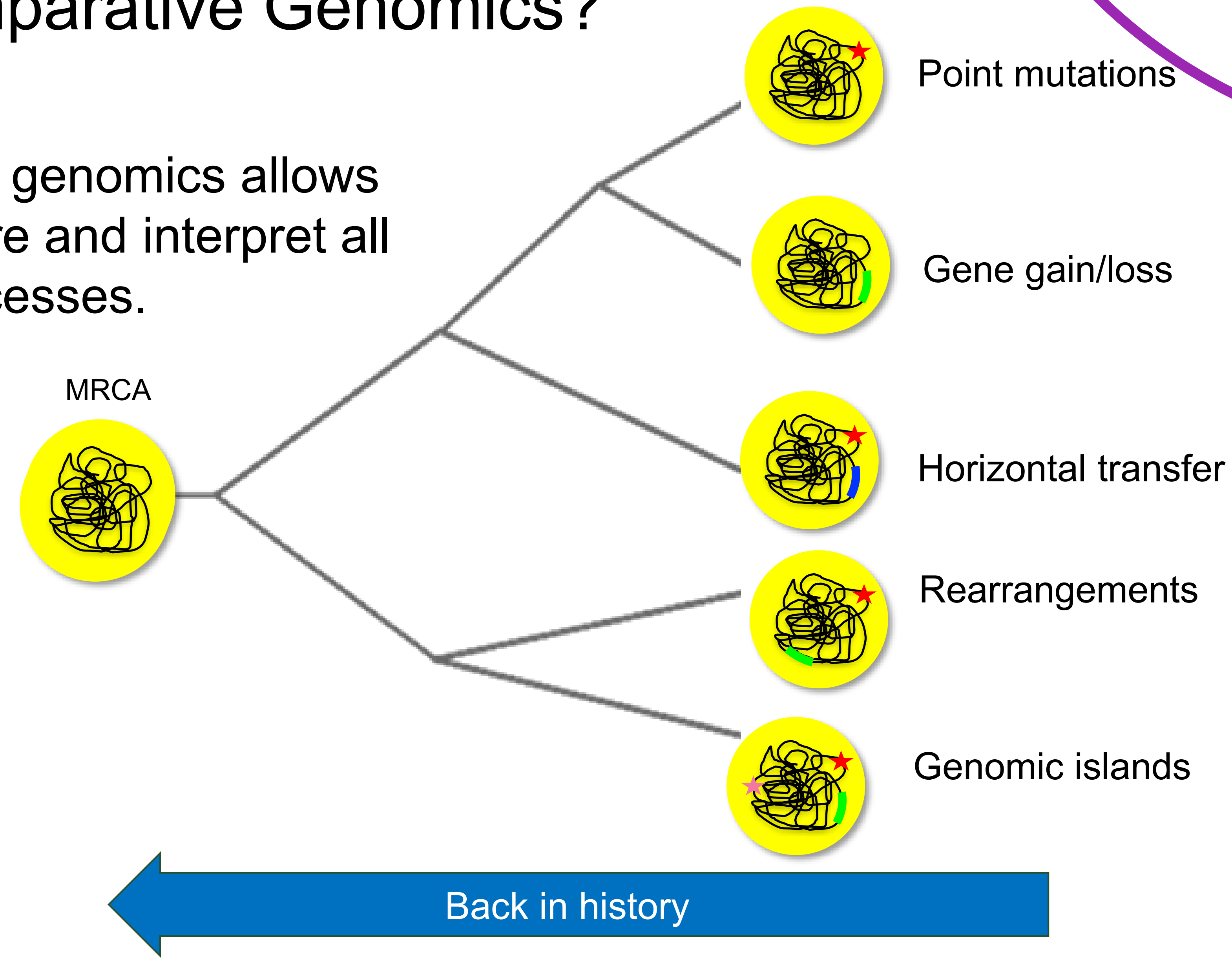
Capture events from the past



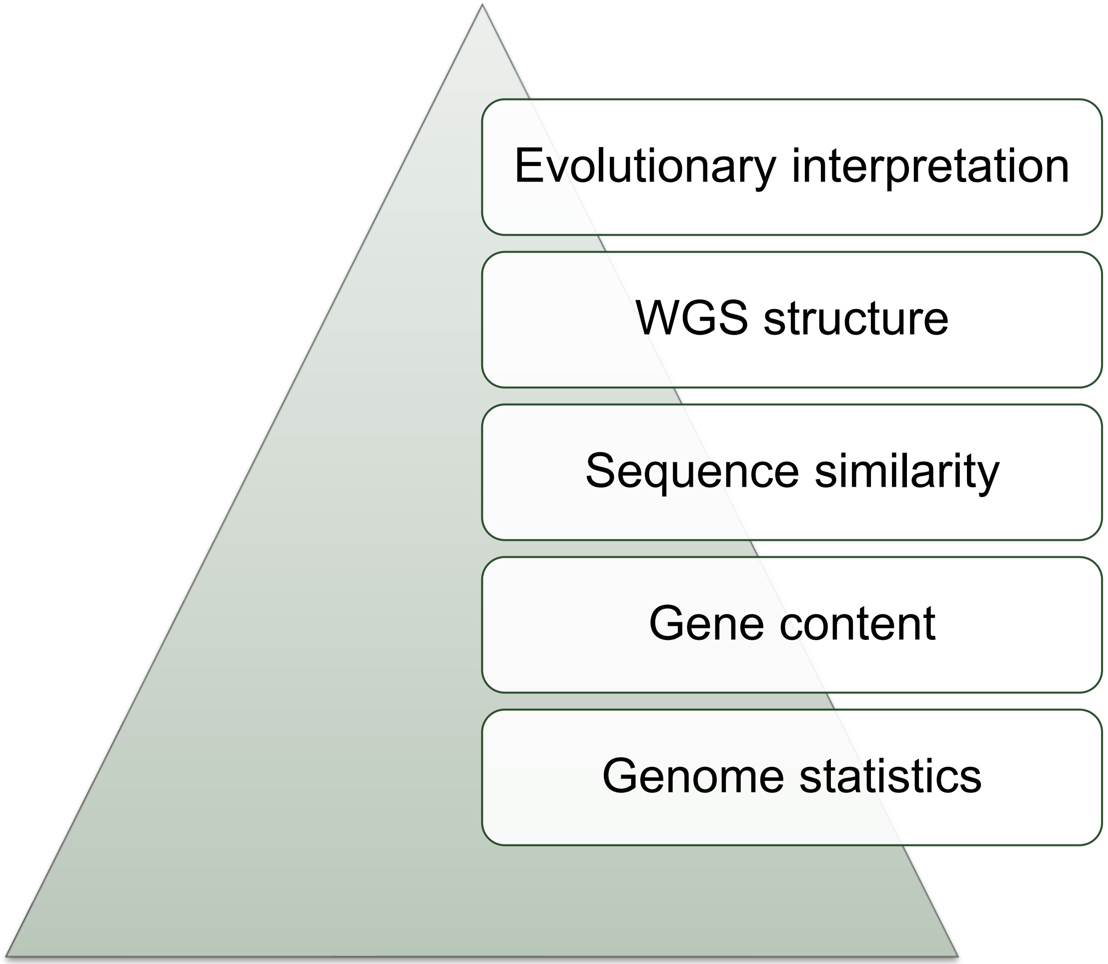
- Impact of selective pressures
    - *Introduction of new antibiotics*
- Adaption to new niches
    - *Host jumps*
- Successful spread
    - *Epidemics expansion*
- Record of their exposure to other bacteria
    - *Mobile genetic elements*
- Record of their exposure to predators
    - *CRISPR spacers*

Why Comparative Genomics?

Comparative genomics allows us to measure and interpret all of these processes.



Levels and Types of Genome Comparison



- Phylogenetics
- Recombination
- Pangenome
- Resistome
- Mobileome
- Phenotype-genotype
- Genome evolution

Basic Genome Metrics

NIH National Library of Medicine
National Center for Biotechnology Information

Search NCBI ...

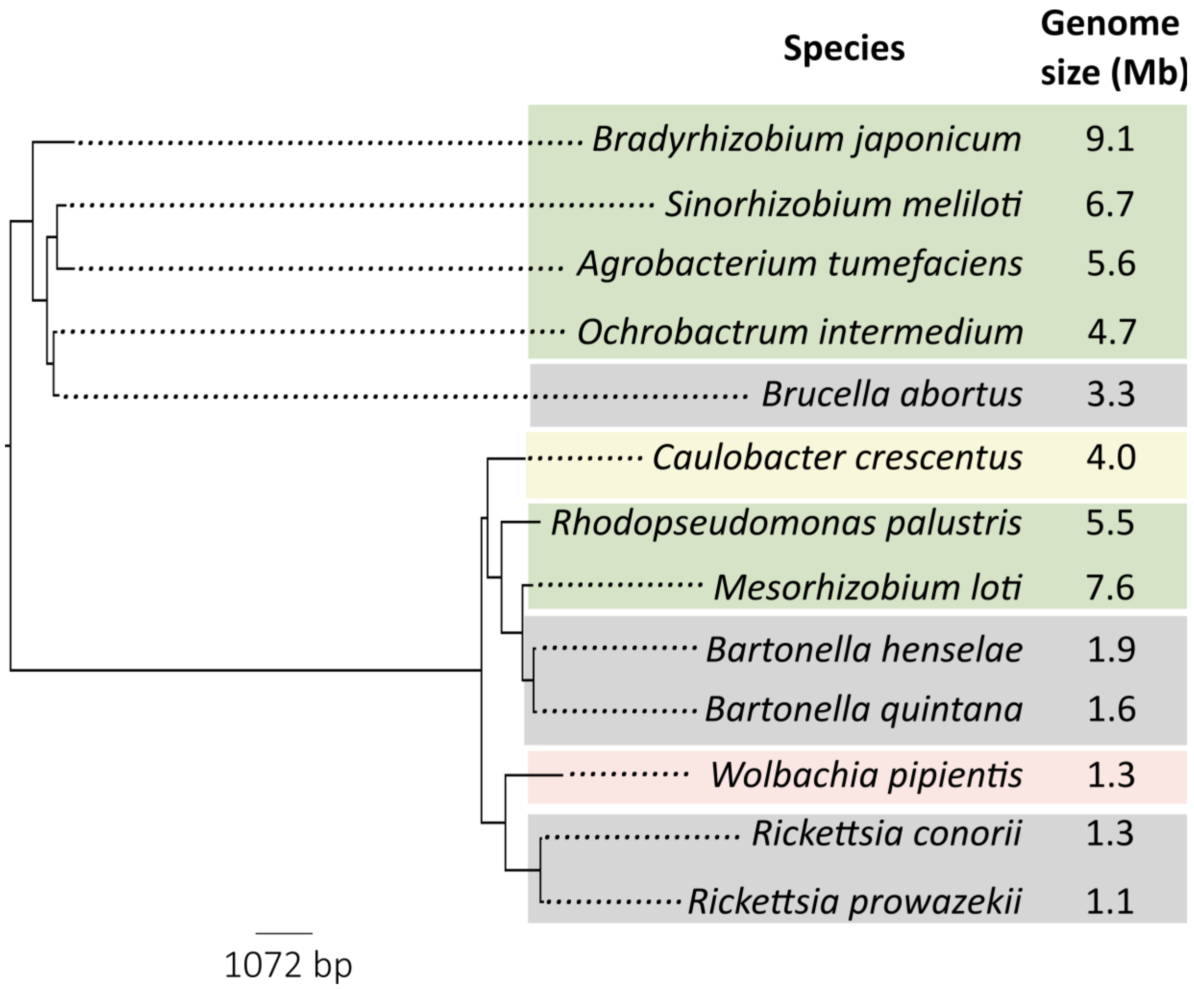
Log in

NCBI Datasets | Taxonomy | Genome | Gene | Command-line tools | Documentation

Genome

Scientific name	Size (Mb)	Chromosomes	GC percent	Protein-coding
Salmonella enterica subsp. enterica serovar Typhim...	4.951	2	52	4,554
Bacillus subtilis subsp. subtilis str. 168	4.216	1	43.5	4,237
Mycobacterium tuberculosis H37Rv	4.412	1	65.5	3,906
Acinetobacter baumannii	3.999	3	39	3,682
Bacillus velezensis FZB42	3.919	1	46.5	3,681

Genome Size Variation



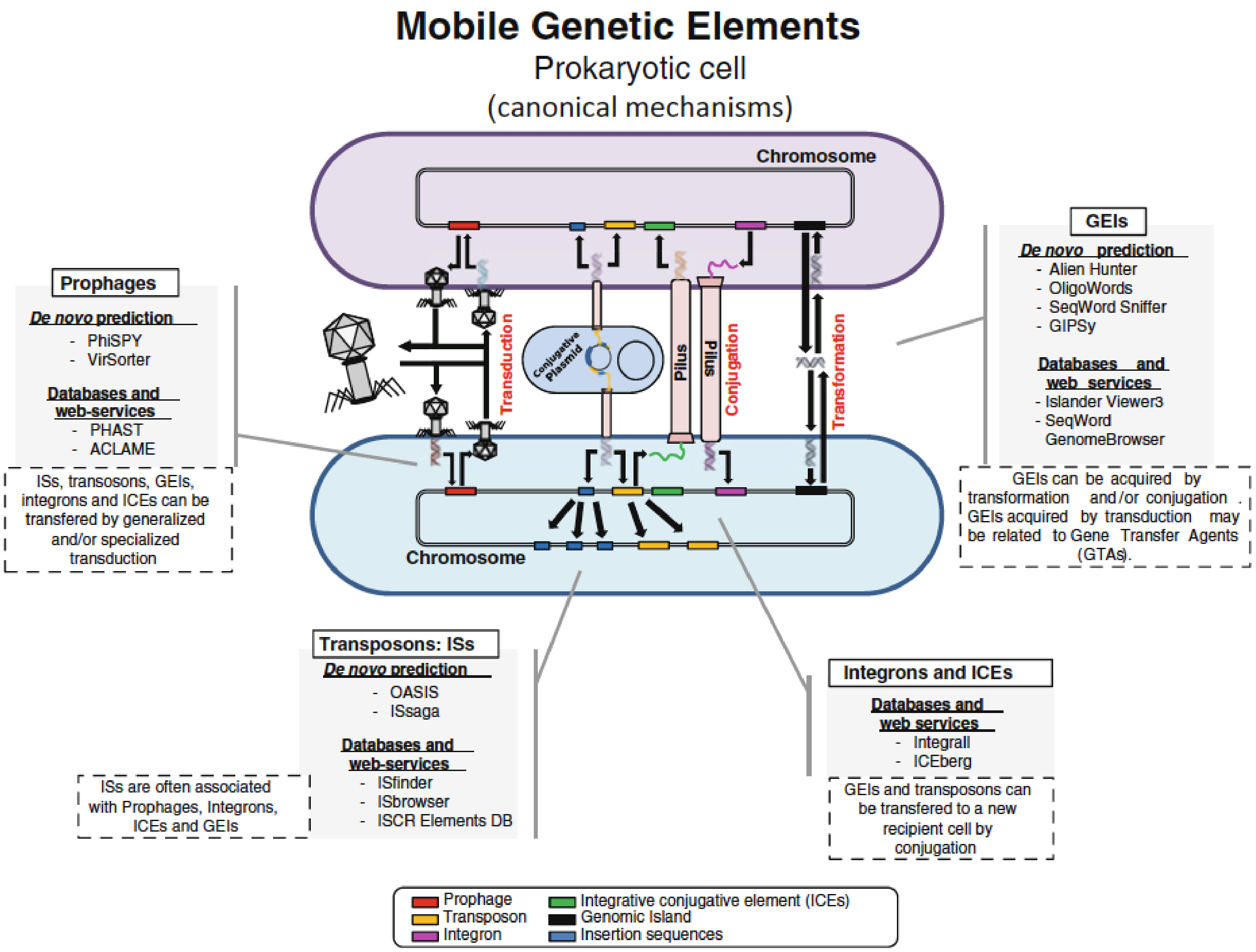
Causes of genome expansion:

- Plasmids
- Phages
- Genomic islands
- Gene duplication

Causes of reduction:

- Host adaptation
- Obligate intracellular lifestyles

Repeats and Mobile Elements

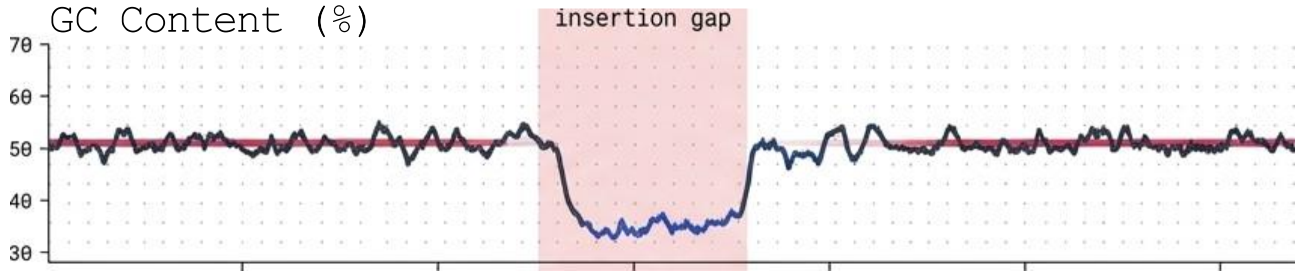


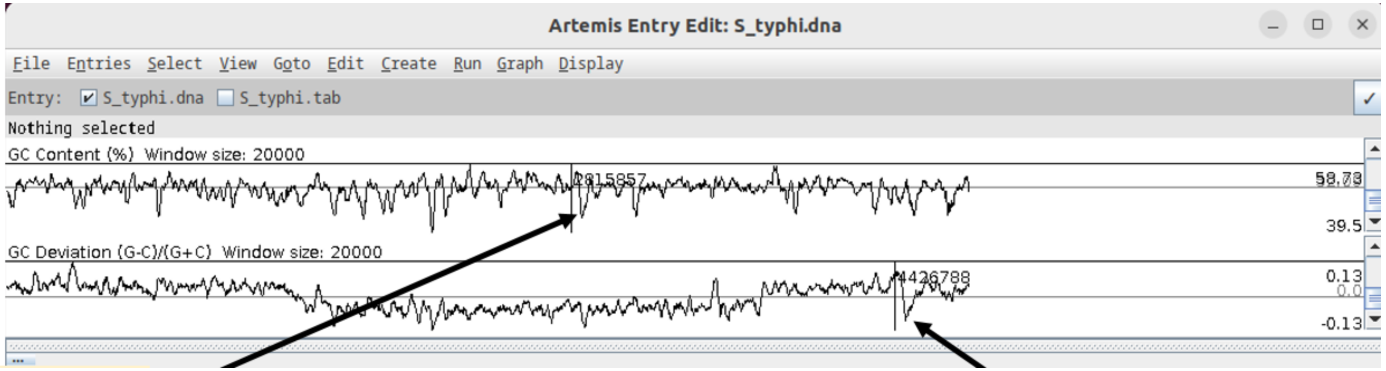
Oliveria-Alvarenga, et al. 2018. doi.org/10.1007/978-1-4939-7463-4_7

GC Content

Why it matters?

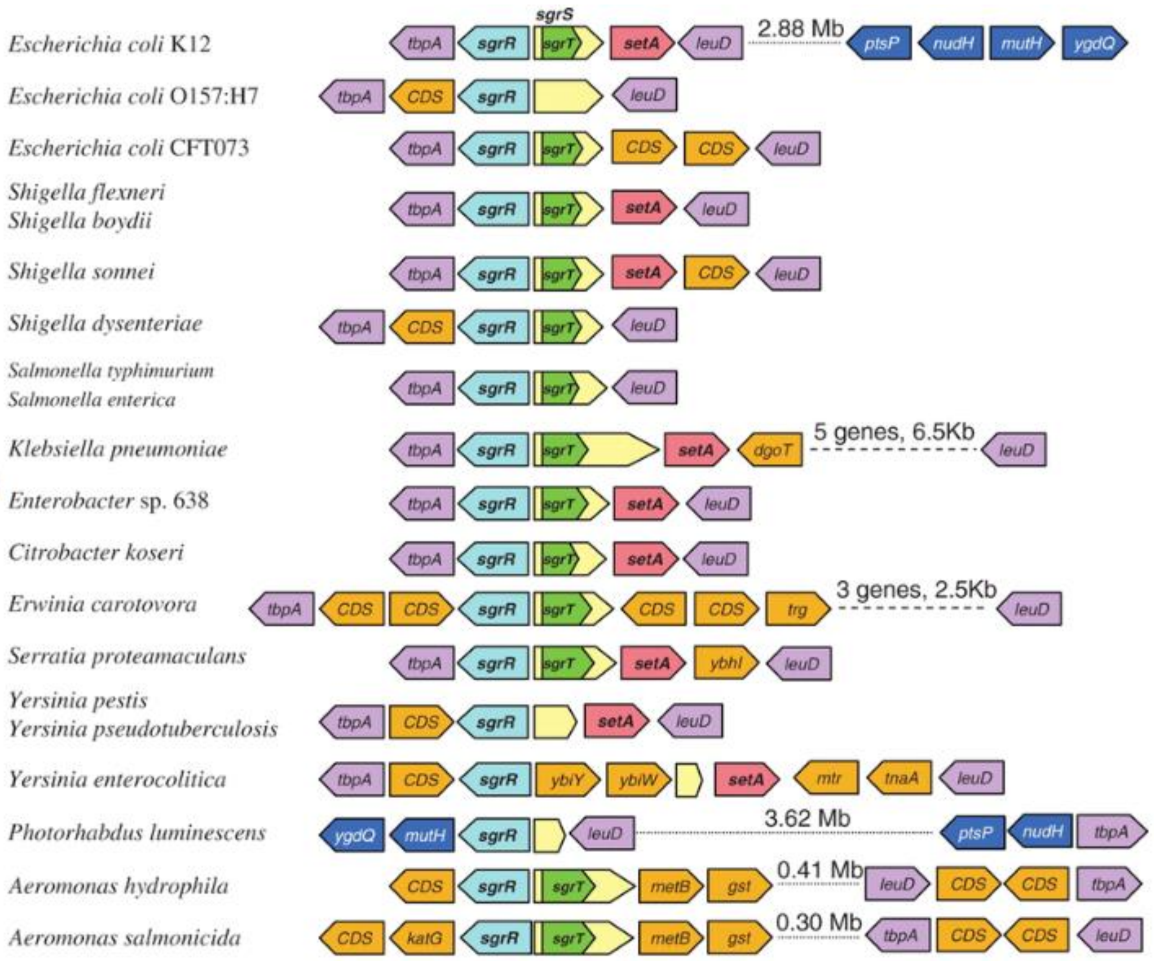
- Species signatures
- Genome stability
- Horizontal gene transfer





Gene Content Comparison

- Which genes are shared?
- Which genes are unique?
- Which pathways differ?

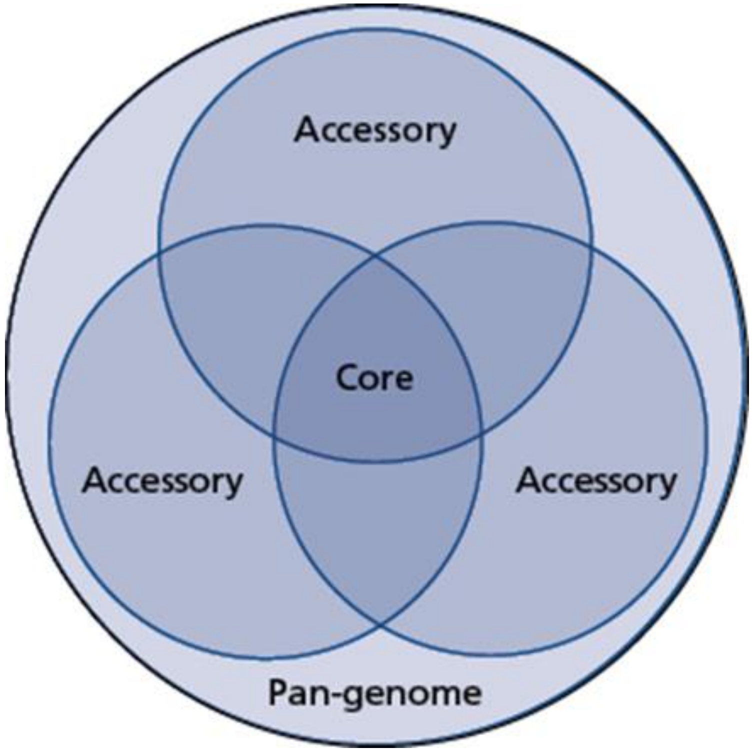


Horler & Vanderpool. 2009. doi: 10.1093/nar/gkp501

The Pangenome Paradigm

The Core Genome

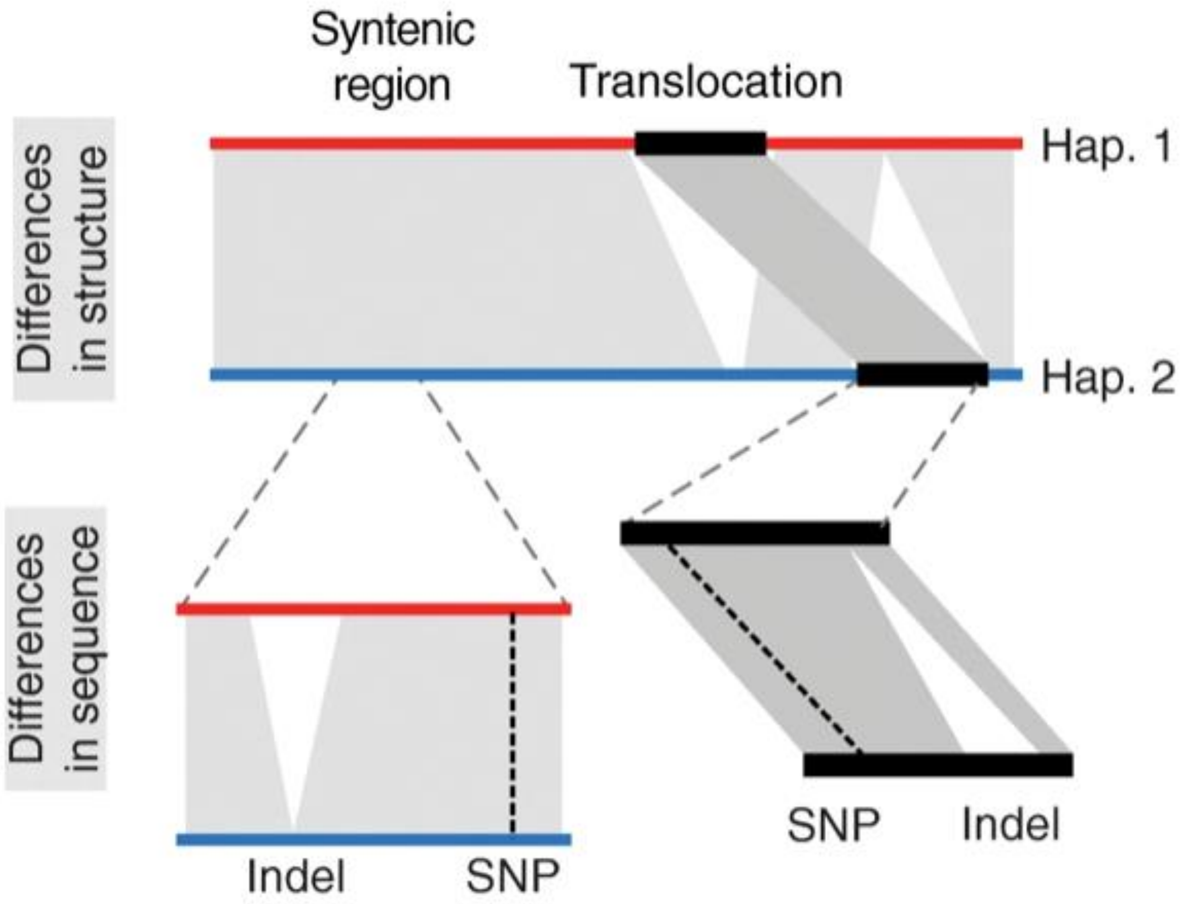
- Conserved sequence shared across a lineage.
- Drives deep phylogeny and high-resolution strain clustering (SNP / cgMLST).
- Highly stable, low mutation rate, vertically inherited



The Accesory Genome

- Variable gene content.
- Drives pathogenicity, environmental adaptation, and outbreak differentiation.
- Highly mobile, horizontally acquired (plasmids, prophages)

Synteny: Looking Beyond Gene Content

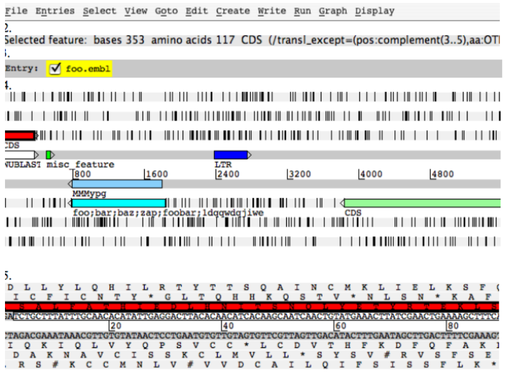


Goel, et al. 2019. *Genome Biol*. doi.org/10.1186/s13059-019-1911-0

The Artemis Suite: Visualizing the Genome

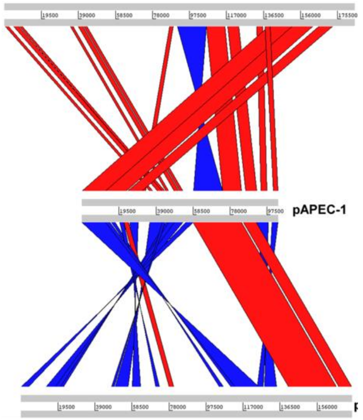
Artemis

Base-level sequencer viewer and annotation editor.
Enables 6-frame translation, GC skew plotting, and real-time feature creation.

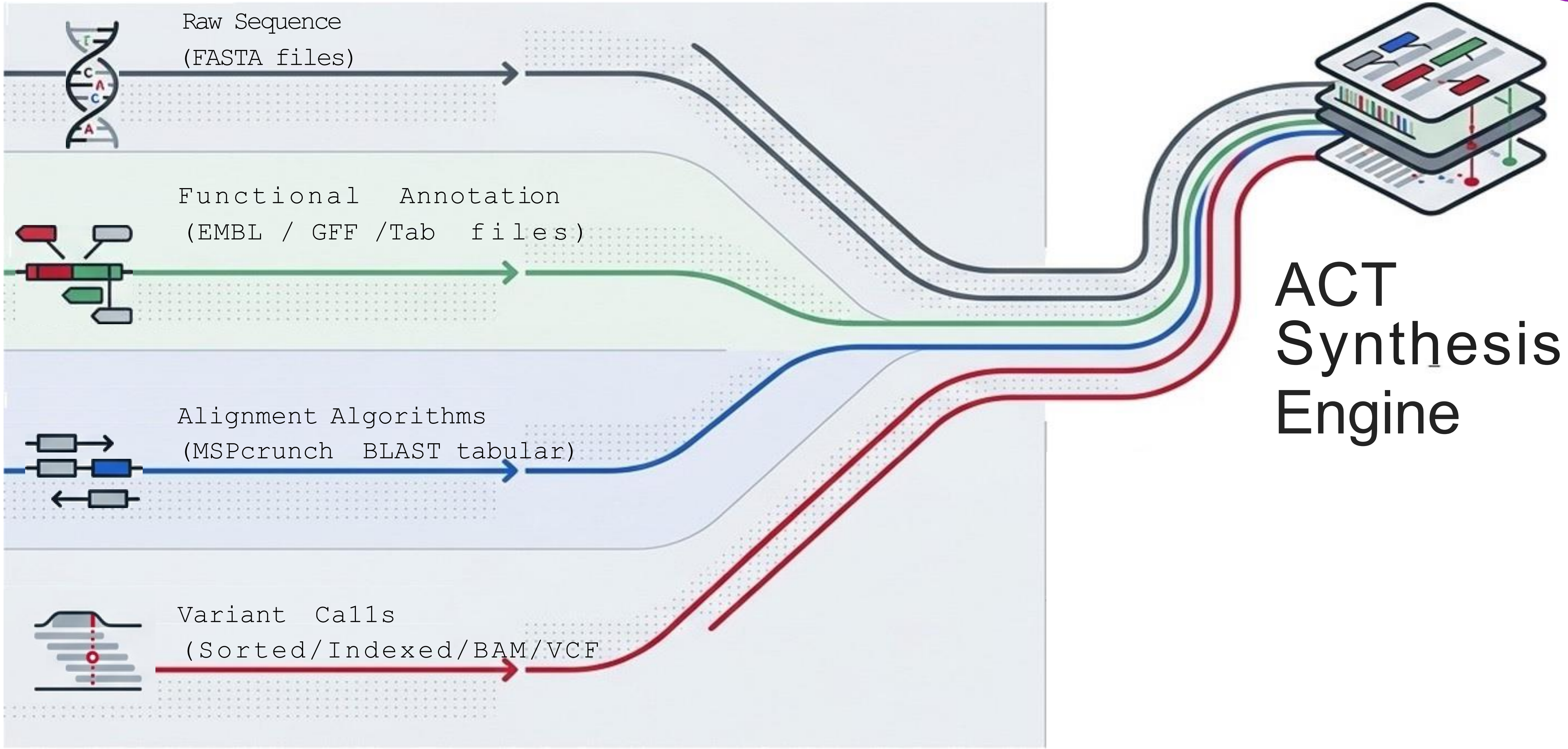


ACT (Artemis Comparison Tool)

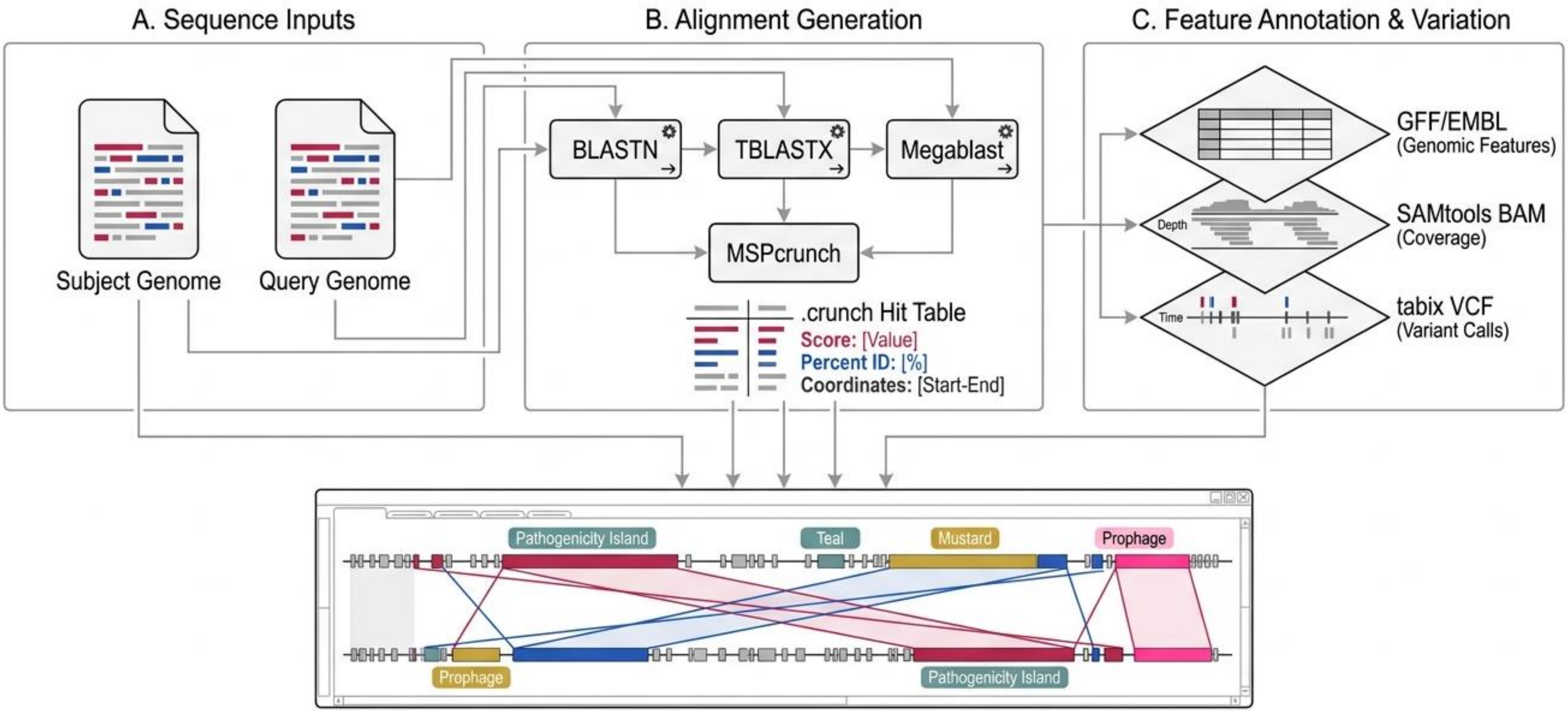
Engine for pairwise or multi-way sequence comparison. Visualizes synteny, insertions, and structural variations.



Data Convergence into the Visual Engine



Pipeline and Integration

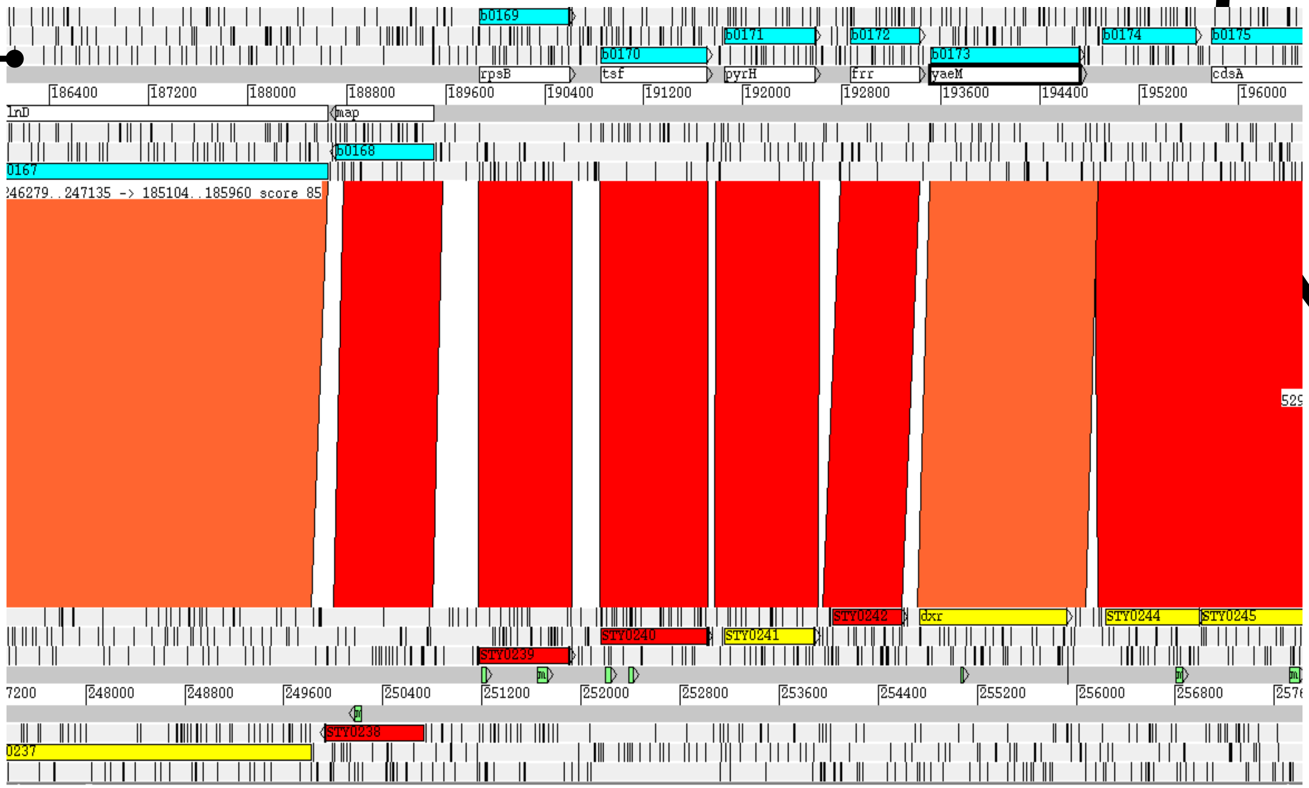


Interpreting ACT Architecture

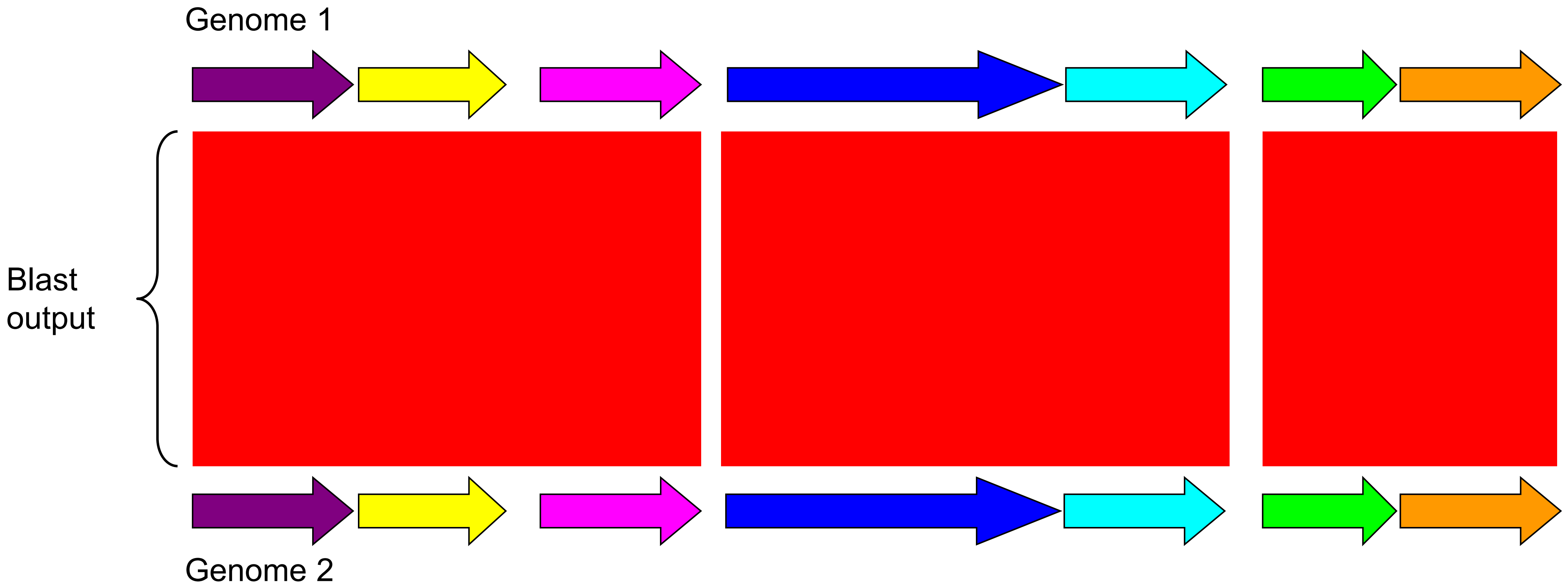
Subject: established reference genome

Query: genome being investigated

Middle layer (comparison view):
- Conserve regions (synteny).
- Inverted regions (reverse complement homology).
- Unique regions (insertions, deletions).

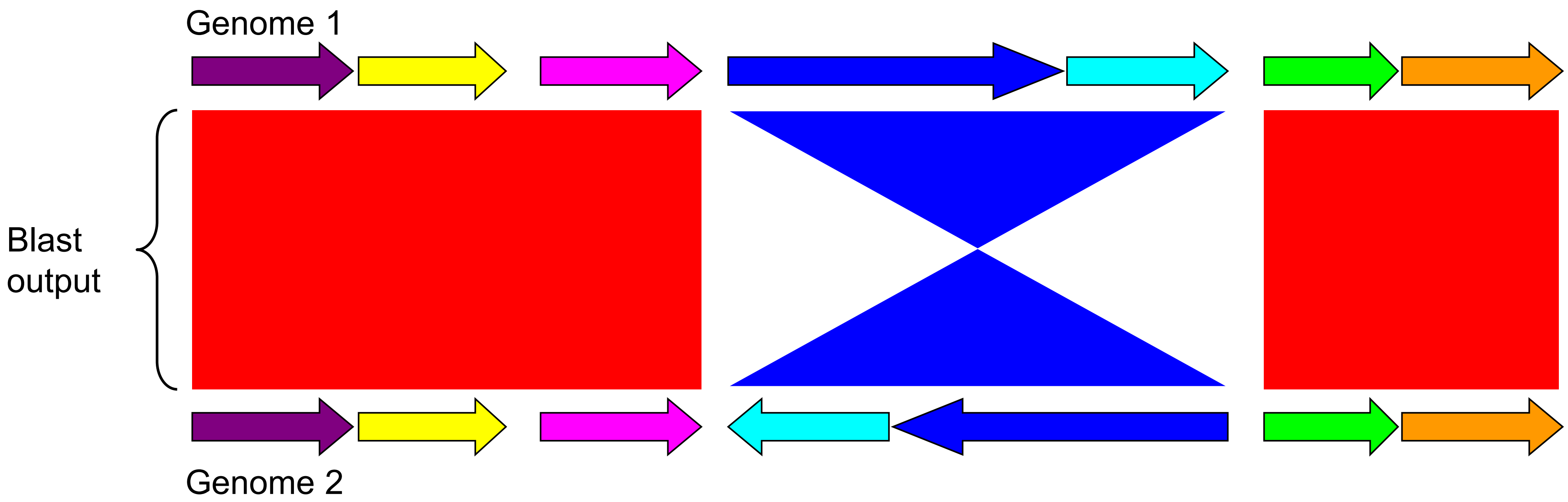


ACT – Artemis Comparison Tool

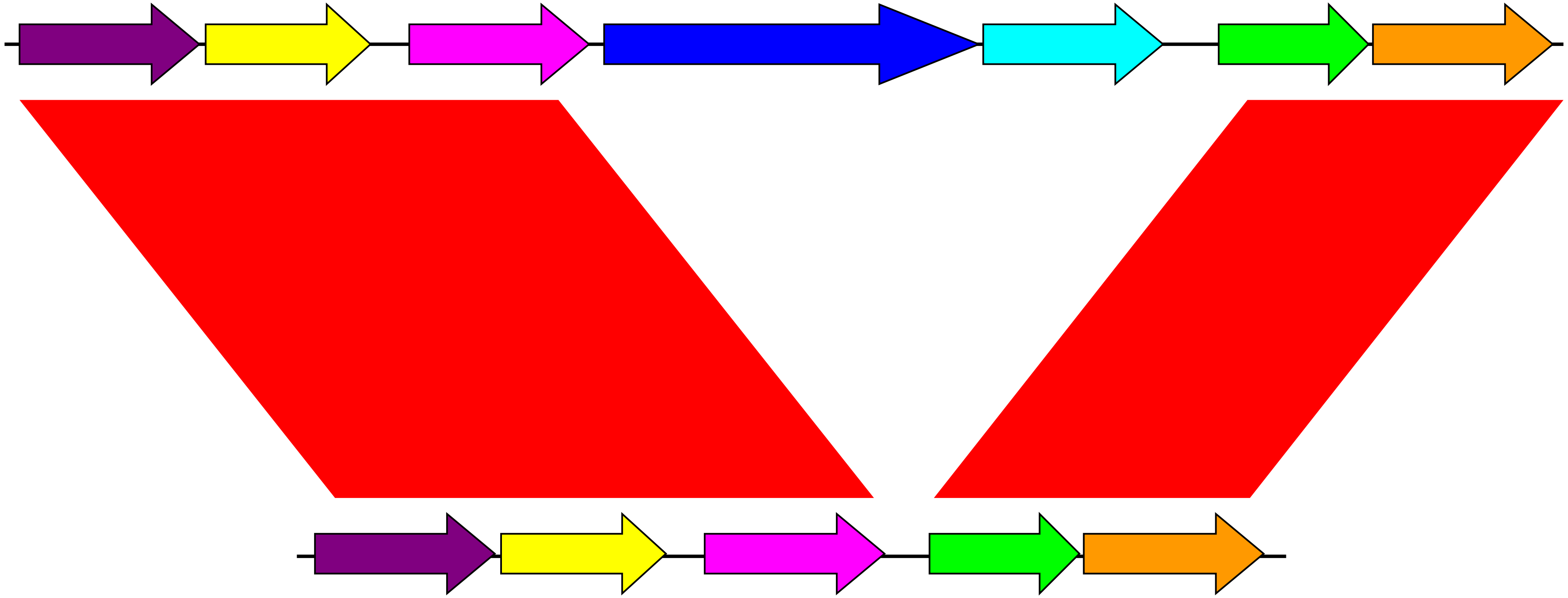


Region of Synteny

ACT – Artemis Comparison Tool

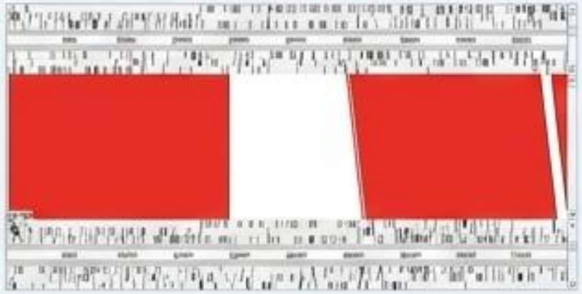


ACT – Artemis Comparison Tool



Example of an insertion or deletion

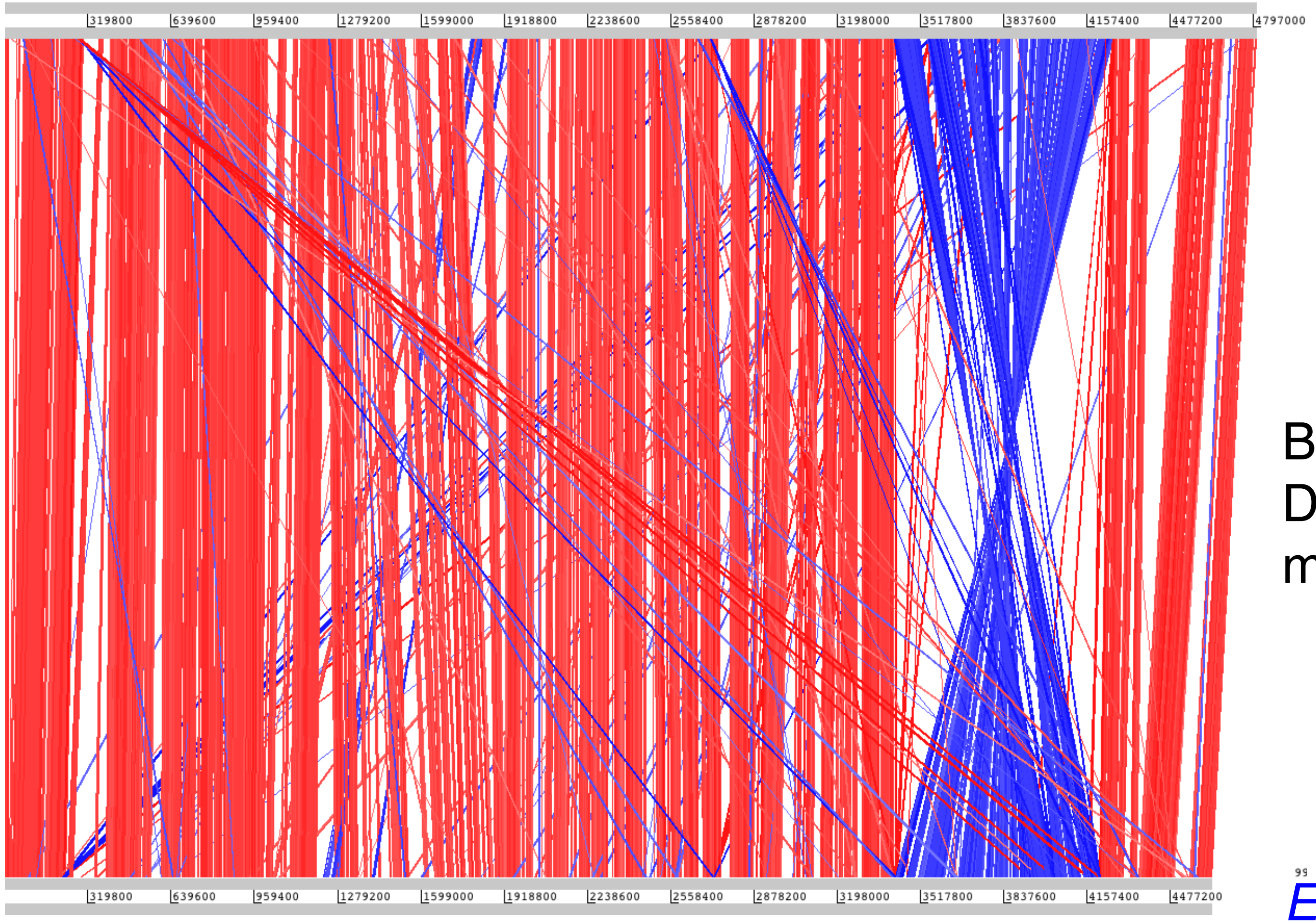
The Synteny Lexicon and Structural Signatures

	Visual Signature	Structural Event	Biological Implication
A		 Conserved Homologous Backbone	Lineage stability and core genomic conservation.
B		 Large-scale Inversion	Homologous recombination (often across inverted repeats).
C		 Insertion / Deletion (Indel)	Horizontal Gene Transfer, gene loss, or acquired pathogenicity islands.

Standard blastall output

```
BLASTN 2.2.6 [Apr-09-2003]


Reference: Altschul, Stephen F., Thomas L. Madden, Alejandro A. Schaffer,
Jinghui Zhang, Zheng Zhang, Webb Miller, and David J. Lipman (1997),
"Gapped BLAST and PSI-BLAST: a new generation of protein database search
programs",  Nucleic Acids Res. 25:3389-3402.

Query= all_bases
         (45,636 letters)

Database: phiPVL.dna
           1 sequences; 41,401 total letters



                                                                 Score    E
Sequences producing significant alignments:                      (bits) Value

all_bases                                                            1.399e+004   0.0

>all_bases
          Length = 41401

 Score = 1.399e+004 bits (7059), Expect = 0.0
 Identities = 7648/7839 (97%), Gaps = 4/7839 (0%)
 Strand = Plus / Plus


Query: 30123 caatagtgttaaatctattttcactaatatgaaaaactggttatctagtacgtggaataa 30182
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct: 11441 caatagtgttaaatctattttcactaatatgaaaaactggttatctagtacgtggaataa 11500


Query: 30183 tatcaaaagcaataccgtcggcaaggctcattcgttatttacaggtgtaaggtctaaatt 30242
             ||||||||||||||||||||||||||||||||||||||||| ||||||||||||||||||
Sbjct: 11501 tatcaaaagcaataccgtcggcaaggctcattcgttatttacgggtgtaaggtctaaatt 11560
```

Chromosomal rearreangemets: *Salmonella*

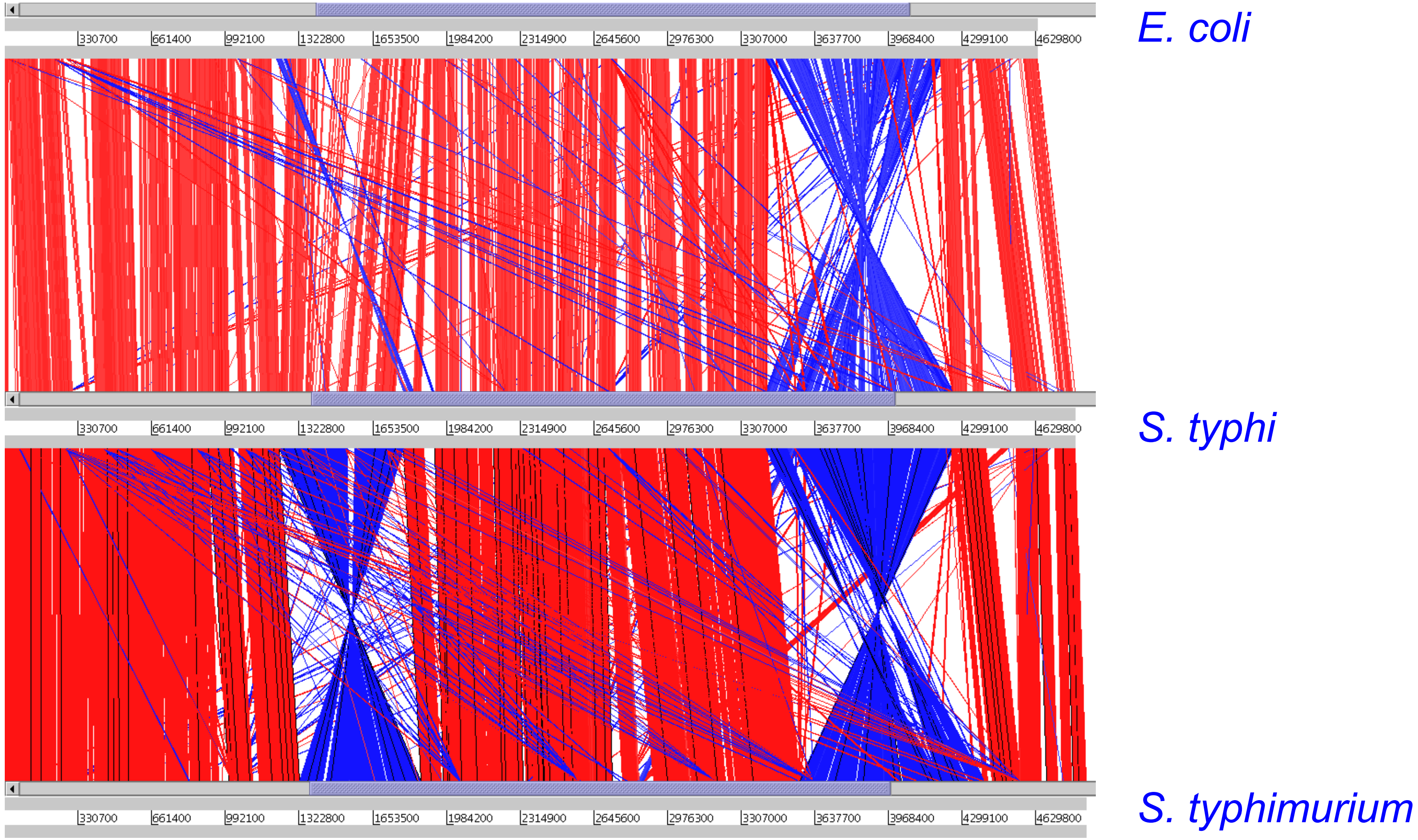


S. typhi

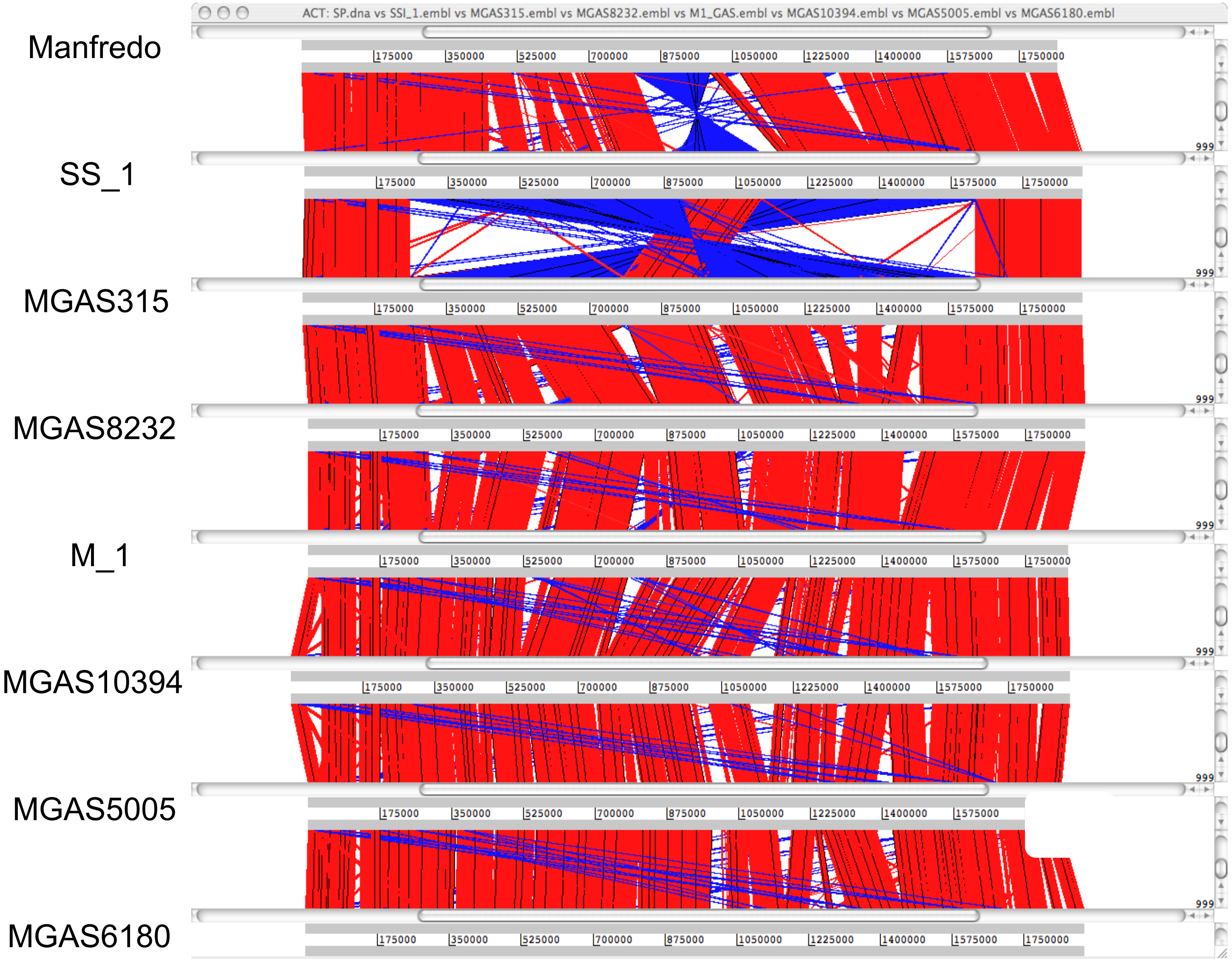
BLASTN
DNA-DNA
matches

E. coli

Chromosomal rearreangemets: *Salmonella*

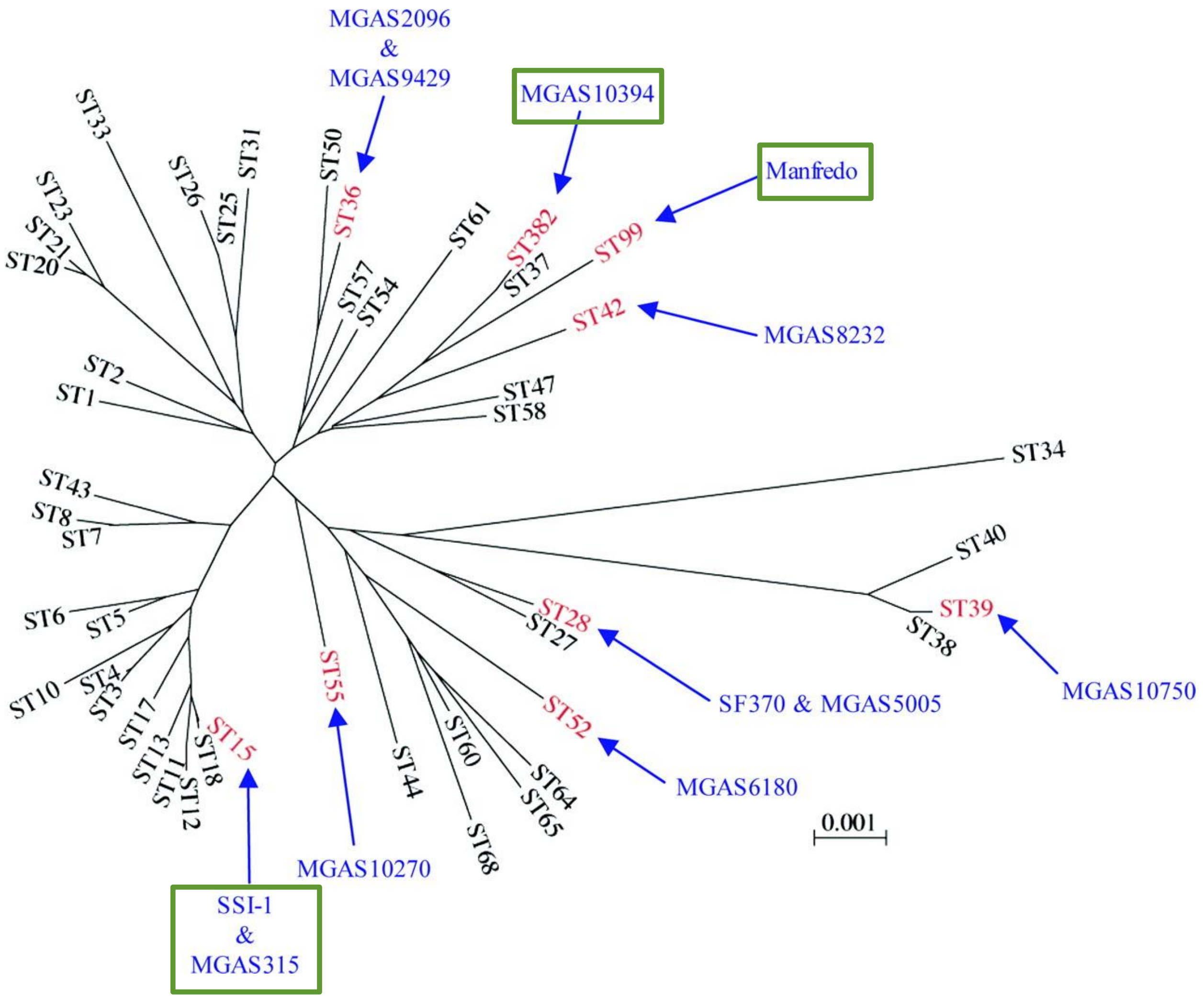


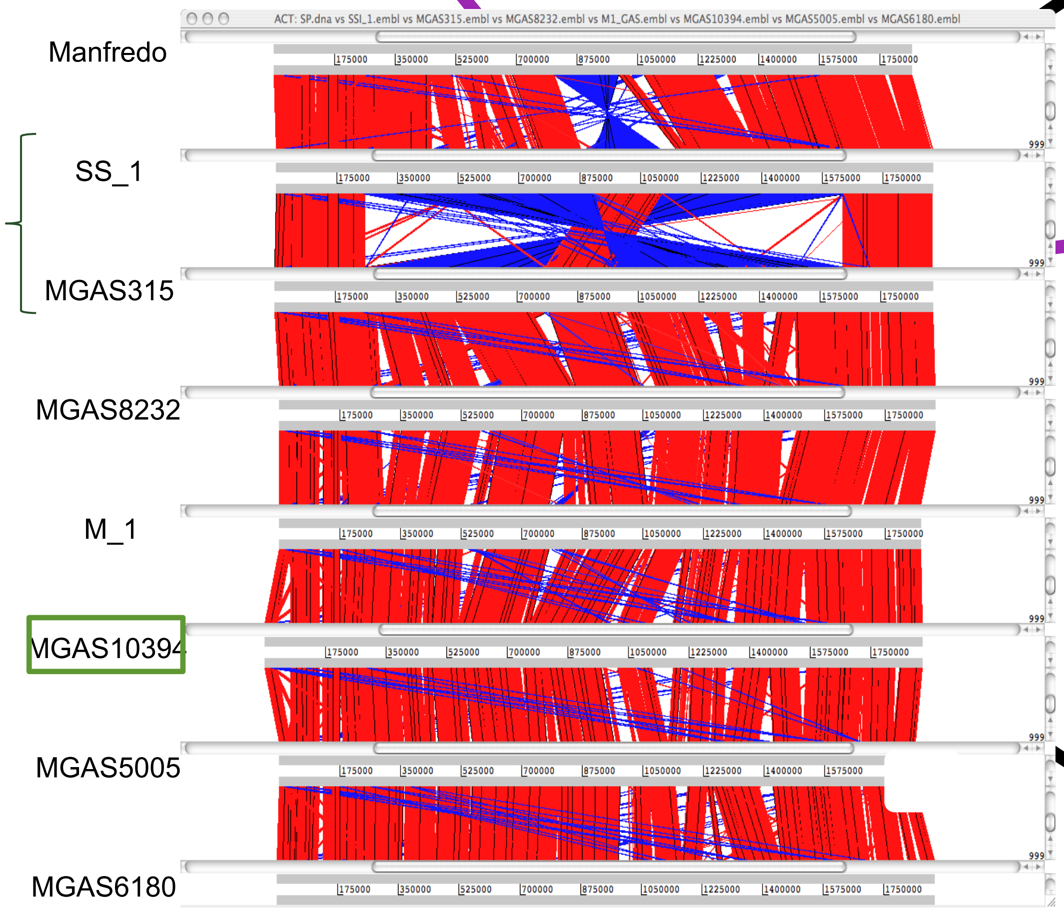
Streptococcus pyogenes comparison



- Diversification by bacteriophages
- Similarities with strains associated with rheumatic fever

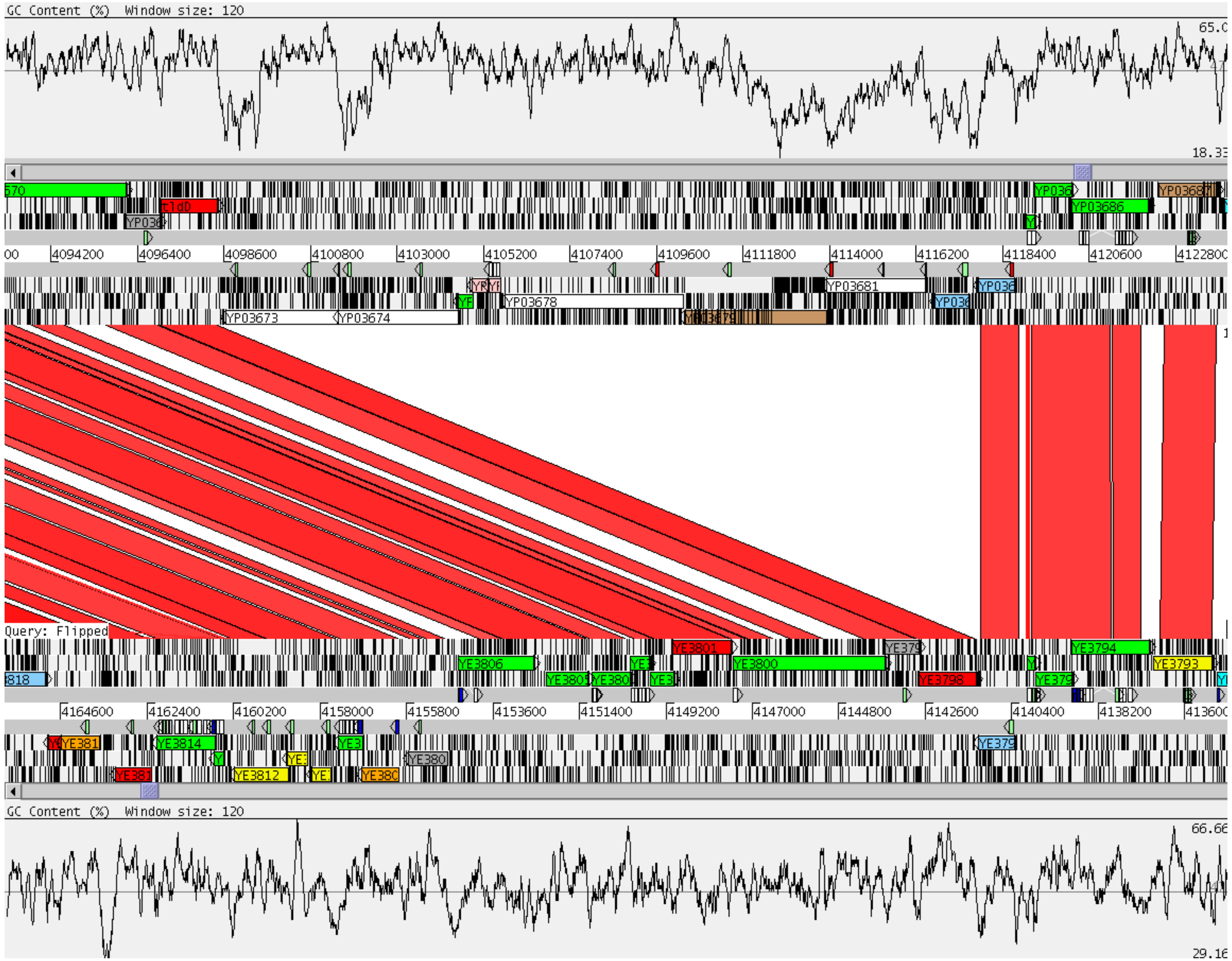
Holden et al. J Bacteriol. 2007 doi: 10.1128/JB.01227-06.





Holden et al. J Bacteriol. 2007 doi: 10.1128/JB.01227-06.

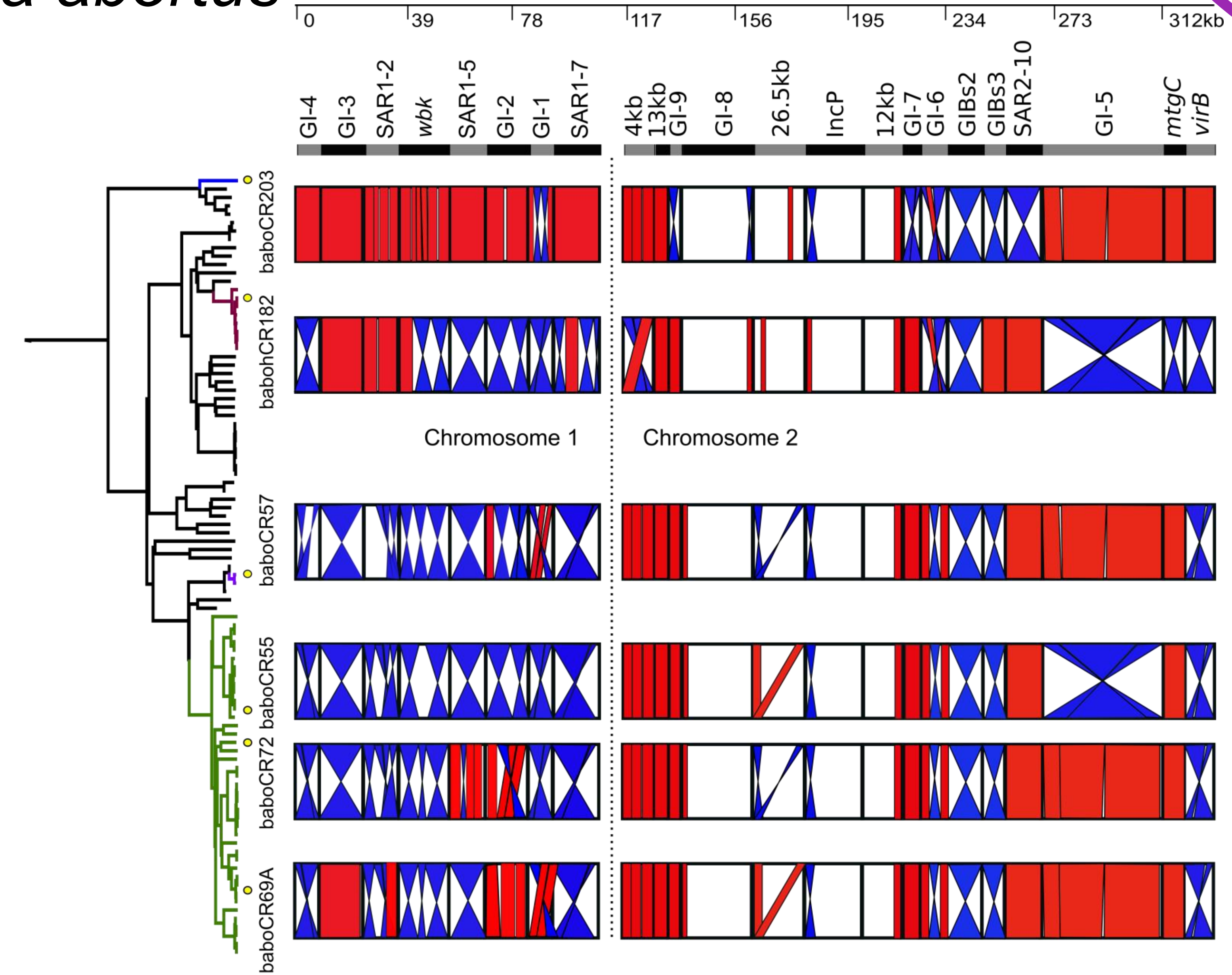
Gene acquisition by *Yersinia pestis*



Y. pestis

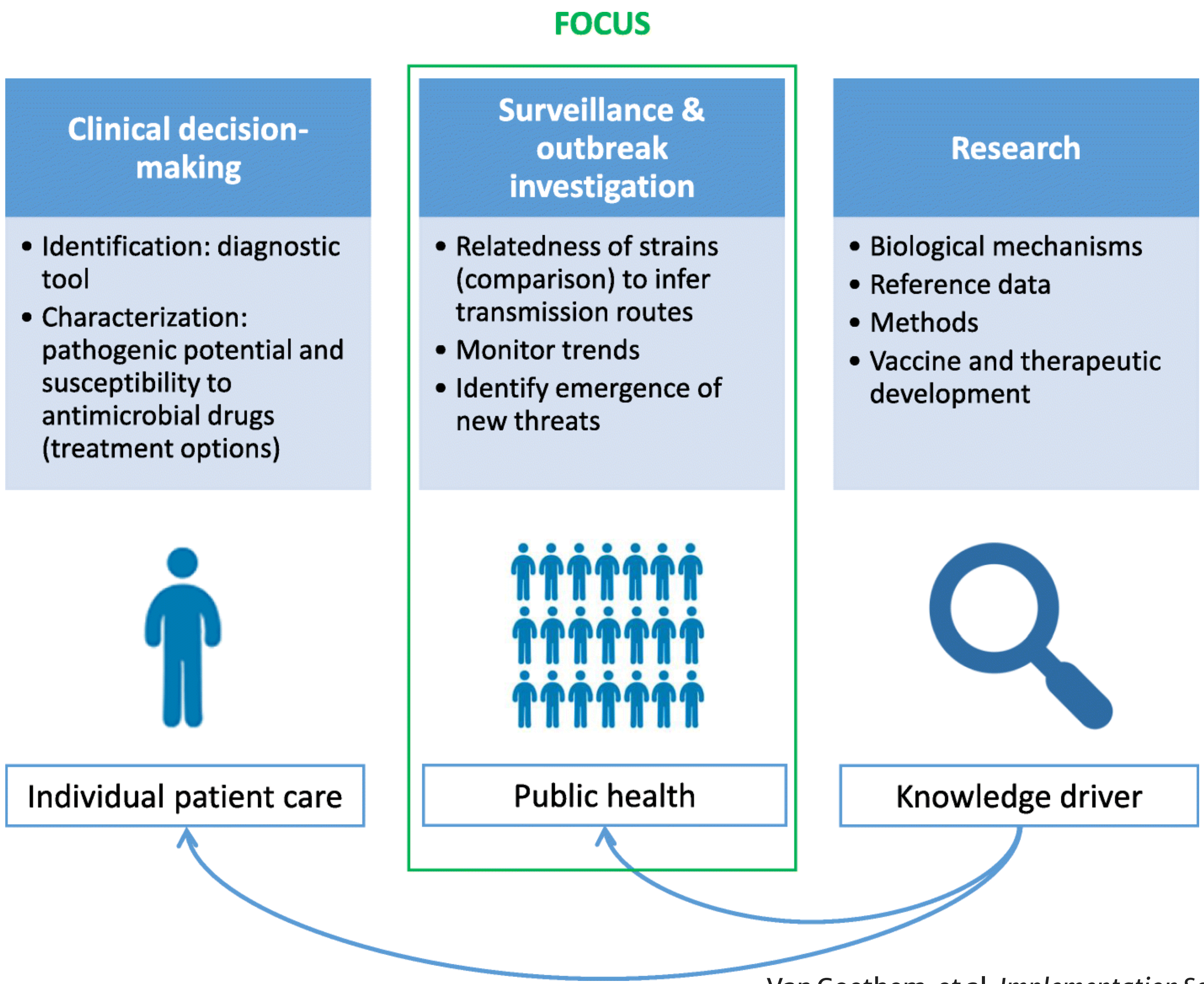
Y. enterocolitica

Brucella abortus



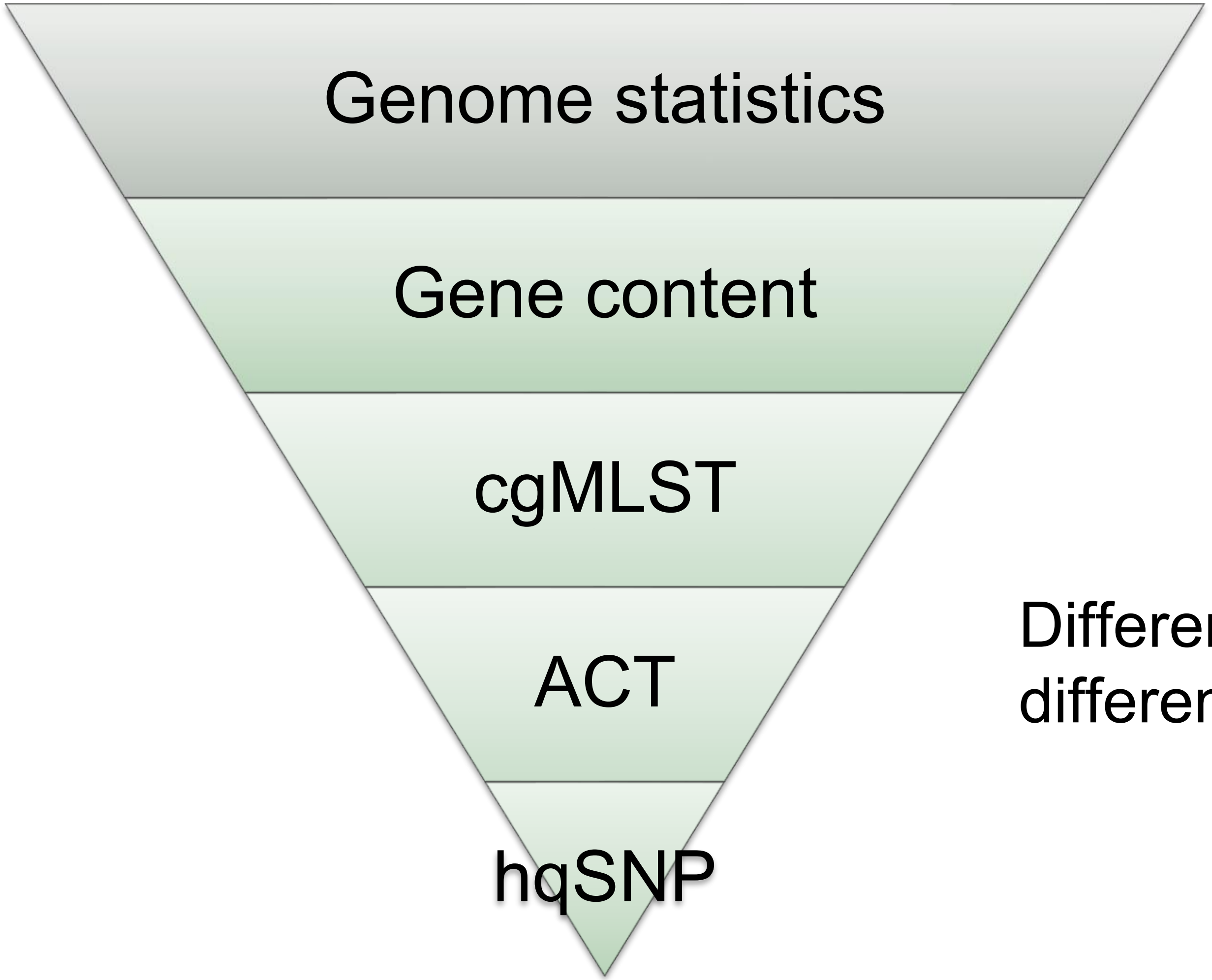
Oviedo-Sánchez *et al.*, 2019, Unpublished

Comparative Genomics in Public Health



Van Goethem, et al. *Implementation Sci*. 2019. doi: 10.1186/s13012-019-0930-2

Different Scales of Comparison

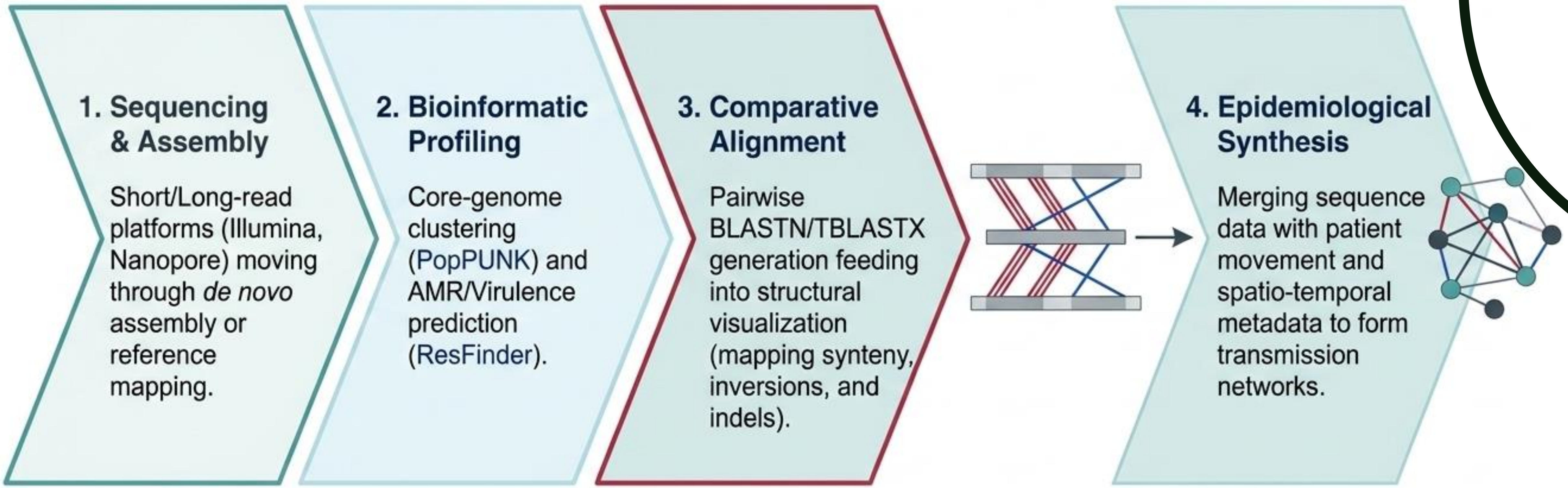


Different questions require different approaches.

It depends !

Method	**Main biological question answered**
Genome size	Has the genome expanded or contracted?
GC content	Is there evidence of foreign DNA or compositional bias?
Gene content	What functions are present or absent?
Pan-genome	How diverse is the species gene repertoire?
Chromosomal alignment	How has genome architecture changed?
SNP analysis	How closely related are the isolates?

Workflow from Clinical Isolate to Public Health Intervention



Why “ACT” Still Matters

Modern surveillance often focuses on:

- SNPs
- cgMLST

But chromosomal alignment reveals:

- Structural variation
- Mobile elements
- Large rearrangements that SNP analyses may overlook.

Take Home Messages

1. No single comparison captures all genomic differences.
2. Genome statistics reveal broad evolutionary patterns.
3. Whole-chromosome alignment provides a powerful view of genome architecture and structural evolution.

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Comparative genomics - Dublin 2026

Module Lead: Marcela Suárez-Esquivel

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Thank you!

Dr. Matt Holden